
LINEAR MODELS : Graded TP

In this TP, we explore the use of the bootstrap method for constructing confidence intervals in linear regression, the FORWARD SELECTION procedure based on hypothesis testing for feature subset selection and model limitations.

Decisions made during the exercises need to be justified. Emphasis is placed on understanding and interpreting the methods; explanations and discussions are expected (code alone receives 0 points).

You have free access to the internet and class materials are allowed.

EXERCICE 1. (Bootstrap for CI.) Bootstrap is a general resampling technique that based on which we can compute estimators, their bias, variance or confidence intervals (CI), ... Besides linear regression, we can find it in random forests or neural networks. In this exercise, we simulate the CI of the regression coefficient using the bootstrap method.

Loading and computing :

- 1) Load the dataset `data1.csv` where the last column is the observation y and the rest is the design matrix X . Verify that the ordinary least squares (OLS) solution using the analytical formula $\hat{\theta} = (X^T X)^{-1} X^T y$ and the `scikit-learn` implementation on the full dataset yield identical results.
- 2) Compute the CI for the slope using the bootstrap method. First, decide and justify the number of bootstrap samples and their size. Then, perform the bootstrap resampling and estimate the regression coefficient for each sample.
- 3) Plot, on the same figure : (i) the original dataset, (ii) the bootstrap regression lines (in a transparent color), and (iii) the regression line obtained from `scikit-learn` on the full dataset.

About the bootstrap samples :

- 4) The *out-of-bag sample* β_i refers to the subset of original dataset not included in the bootstrap sample B_i . Explain how these observations can be used for cross-validation.
- 5) Show that the probability that a specific observation (x_j, y_j) appears in a bootstrap sample is $1 - e^{-1}$.
- 6) Verify this result empirically : report the number of distinct observations in each sample.

Analysis of results :

- 7) Plot a histogram of the bootstrap slope estimates and highlight the corresponding bootstrap CI on the same graph. Does the true slope coefficient (which is $\theta_1^* = 2$) fall inside this CI?
- 8) Compute the asymptotic CI. Does the true slope coefficient fall inside the asymptotic CI?

EXERCICE 2. (Feature subset selection.) This exercise concerns the feature subset selection method introduced in the last class (see details in the slides and the lecture notes) based on hypothesis testing of the null effects of the coefficients. Only `scikit-learn`, `numpy` and `pandas` are allowed (`statmodels` or similar are NOT allowed).

In this exercise, we apply the FORWARD SELECTION procedure to the explanatory variables. At each iteration, we select the *most relevant variable* for prediction. To decide which variable is the most relevant, we perform a "test of no-effect" to check whether $\theta_j^* = 0$.

In the first part of the exercise, we proceed as follows :

- 9) Load the data `data2.csv` where the last column is the observation y and the rest is X . Perform the forward selection with no stopping criterion : At each iteration, perform the hypothesis test, decide which variable to select, and store the p -value of the selected variable. There is no stopping criterion yet (this will be introduced later), so continue until all variables have been selected. Provide the ordered list of selected variables.
- 10) Justify in the notebook the hypothesis of the test, the test statistic, and the computation of the p -value.

Stopping criterion. We now improve the previous version by introducing a stopping rule : the procedure stops when the corresponding p -value exceeds the threshold 0.05. Perform the same experiment with this stopping criterion and illustrate the behavior of the algorithm by :

- 11) plotting the three graphs of the statistics obtained from the selection of the 1st, 2nd, and 3rd variables ;
- 12) plotting the first 50 p -values (each associated with a selected variable), and drawing a horizontal line at the threshold providing the final list of selected variables.

EXERCICE 3. (Underdetermined system.) In this exercise we explore linear regression in a dataset where standard methods may fail.

- 13) Load the dataset `data3.csv`, where the last column corresponds to the response y and the remaining columns form the design matrix X . Compute the OLS solution using the analytical formula $\hat{\theta} = (X^T X)^{-1} X^T y$ and compare it with the `scikit-learn` implementation. Do they produce identical results ? Describe any issues you observe and state the cause of the problem.
- 14) We have seen two methods to overcome this limitation. Implement both and discuss which one produces results closest to the `scikit-learn` solution.